

Determinants of Restructured Assets in Indian banks

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Abstract

This paper investigates the determinants of restructured advances in Indian commercial banks from 2004-2005 to 2015-2016. The major AQR by RBI in 2015 revealed the true picture of stressed assets in banks and it became evident that banks were hiding NPAs under the practice of regulatory forbearance. An important aspect of this rising stressed assets was the rise in restructured advances in banks which has often been ignored despite being a crucial factor for the rise in NPAs. This paper fills this void and examines the determinants of these restructured advances in banks. The primary purpose is to find what factors motivate banks to restructure advances?. Using a panel data for Indian banks the study examines the determinants of restructured advances in banks. The study would have implications for bank managers, practitioners and policy makers looking to frame measures to curtail rising NPAs in banks.

Keywords: Indian commercial banks, Asset Quality Review, stressed assets, regulatory forbearance, restructured assets.

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Introduction

Banking industry forms the essential component of the financial system of a nation. As an important segment of financial sector banks engage in financial intermediation and resource allocation providing a cushion to the financial stability of a country. Banking sector in India is the largest and the most important segment of the financial sector. Indian financial system is mostly a bank based system as commercial banks account for more than 64 percent of total assets held by the financial sector (Indian Brands Equity Foundation, 2016).¹ Despite banking sector reforms and conservative policies by banks, Non-performing Assets (NPAs) and asset quality remain one of the biggest challenges for banks in India. Rajan (2016) highlighted stressed assets, bank governance and liquidity as some of the major challenges for banks in India.² According to Arvind Subramanian Chief Economic Advisor Govt. of India NPAs problem is the number 1 macroeconomic challenge for India.³ Ownership is a very critical component of rising NPAs in Indian banks as galloping bad loans are a severe problem in public sector banks (PSBs) as compared to private sector banks. Govt. of India, RBI and bank managers are actively looking to identify the potential determinants of NPAs so that appropriate policy mix to contain rising stressed assets is implemented. The major Asset Quality Review (AQR) by RBI in 2015 brought the problem to the fore as total stressed assets in PSBs were significantly higher which if not controlled could trigger a banking crisis. A very crucial component of the burgeoning stressed assets was the restructuring of standard advances by banks. A large number of stressed accounts

¹ Source: <http://www.ibef.org/industry/financial-services-india.aspx> accessed on 15th June, 2017.

² Source: <https://rbidocs.rbi.org.in/rdocs/Speeches/PDFs/CIIL11021609A8EB38FFF444DD8461EC3F244FEFA5.PDF> accessed on 10 June, 2016.

³ <http://www.moneycontrol.com/news/business/economy/npa-issue-is-indias-no-1-macroeconomic-challenge-arvind-subramanian-2248985.html>, accessed July 1, 2017.

considered 'standard' in bank books were actually substandard. The RBI's AQR, 2015 showed that banks were using restructuring of assets as a leeway to sweep NPAs under the carpet as it enabled them to maintain lower provisions for these restructured accounts. A large body of literature on NPAs has focused on identifying the determinants of NPAs. As defined by Reserve Bank of India (RBI), "a restructured account is one where the bank grants concessions to the borrowers which would not have been given otherwise. This restructuring would generally involve alteration of terms of advances like repayment period, repayment amount and rate of interest etc. It involves nurturing of viable unit which has been adversely affected."⁴

Literature Review

Sensarma & Chaudhuri (2008) in their study on Indian banks found that the establishment of debt recovery tribunal had a negative and significant impact on bank NPAs in India. Louzis et al., (2012) studied the bank specific and macroeconomic determinants of Nonperforming loans (NPLs) in Greece using the dynamic panel data method from 2003 to 2009. The results present that GDP growth, unemployment, lending rates and public debt had a strong impact on NPLs of banks in Greece. Kauko (2012) connects external deficits with the NPLs of the banks and finds that both credit growth and current account deficits predict the changes in NPLs of banks. Ghosh (2015) explored industry specific and regional economic determinants of NPLs for commercial banks in U.S. from 1984 to 2013. Using fixed effects and dynamic panel estimation author finds out that greater capitalization, liquidity risks, poor credit quality, cost inefficiency and size increases NPLs while greater profitability was associated with lower NPLs. Regarding the regional economic determinants, greater state real gdp, personal income growth, state housing price index reduces non performing loans while inflation, unemployment and public

⁴ <https://rbi.org.in/Scripts/Glossary.aspx> accessed on Oct 10, 2016

debt increases NPLs. Dhar & Bakshi (2015) explored the determinants of bad loans in Indian public sector banks from 2001 to 2005. The regression results show lending to sensitive sectors as a positively significant variable while capital adequacy ratio (CAR) and net interest margin as negatively significant variables affecting NPAs in Indian banks. Us (2016) attempted to study the impact of global crisis on determinants of NPLs in Turkey. The analysis is conducted for pre and post crisis periods from 2002 to 2013 using dynamic panel system GMM estimation. While bank specific factors like lending, capital adequacy, profitability etc. played a crucial role in explaining NPLs pre crisis, post crisis period changes in NPAs were largely explained by macroeconomic and policy factors. Samantaraya (2016) studied the determinants of non-performing for commercial banks in India and found that the past credit growth and behavior led to increase in the current NPAs on account for poor screening and adverse selection problems. Sarkar & Sarkar (2016) in their study on board governance in Indian banks tested the relationship between board characteristics and bank outcomes.⁵ Their findings demonstrate that board independence impacts bank outcome, CEO- Chairman Duality and presence of nominee directors have a negative impact on bank outcomes. Ahamed & Mallick (2017) conducted a study on Indian banks to examine the impact of restructured assets⁶ on bank risk from 2003 to 2012 and found that higher amount of restructured assets by banks in the sample period reduced bank risk taking as banks got benefitted from low provisions. Also, this impact was more pronounced for banks with lower loan loss provisions. Though the extant literature has focused on determinants of NPAs in banks it is void of academic research on determinants of restructured assets in banks. The study fills this void in the literature and examines the determinants of restructured advances in banks making following contributions to the existing literature: First, to

⁵ Bank outcomes include Return on assets, Asset quality etc.

⁶ In this study we refer to restructured standard advances.

the best of our knowledge this is the first study which examines the factor influencing restructured advances in banks. Second, this study adds to the empirical literature on bank NPAs which has predominantly focused on identifying the determinants of NPAs in banks without focusing on restructured advances which are very prominent factor for rise in NPAs in case of Indian banks. Third, policy makers and regulators in India would find this study useful in framing policies for restructured standard advances in Indian banks.

The objective of this paper is to identify bank specific determinants of restructured assets in Indian banks. The paper investigates the association between bank specific variables and restructured assets in Indian banks. The existing literature on determinants of bank NPAs identifies two major groups of determinants as bank specific determinants and macroeconomic determinants which influence bad loans in banks. This paper focuses on bank specific determinants of restructured assets in banks.

Methodology, Data and Hypotheses

The paper uses panel data to investigate the determinants of restructured assets in banks. The study uses data on 49 commercial banks from 2005 to 2016. Cooperative banks, regional rural banks, payment banks and small finance banks are excluded from the study because of the quality concerns. The data on all variables has been retrieved from RBI database statistical tables related to banks in India. The study proposes the following econometric specification:

$$Y_{it} = \beta_0 + \sum_{i=1}^k \beta_k X_{kit} + \mu_{it}$$

The dependent variable is the bank wise ratio of restructured standard advances⁷ as a percentage of gross advances. The independent variables chosen for the analysis are net interest margin, capital adequacy ratio, gross NPAs as a percentage of gross advances, Ownership dummy 1 for PSBs and 0 otherwise. The choice of these variables is based on the existing literature.

A large chunk of Indian banks restructured their stressed accounts so as to reduce NPAs (Ahamed & Mallick, 2017). Bank NPAs is a critical factor influencing restructuring in banks. Indian banks with high NPAs will undergo more restructuring⁸ as this gives them a leeway to maintain lower provisions which eventually increases their profitability. We postulate our first hypothesis as following:

H1- Higher NPAs in Indian banks will be associated with higher restructuring of standard assets.

Bank ownership plays a crucial role in determining changes in restructured assets. Particularly PSBs made use of Corporate Debt Restructuring (CDR) and restructuring mechanisms so as to upgrade non performing restructured assets to standard category after a certain period and charge less to their net income (Ahamed & Mallick, 2017). Ahamed & Mallick (2017) in their study on Indian banks found that PSBs with higher restructuring of assets could reduce their risk more as compared to private and foreign banks. Since, PSBs have greater association to sick corporates, possess higher NPAs, stressed assets we postulate the second hypothesis as following:-

H2- PSBs are associated with higher restructuring of standard accounts as compared to private and foreign banks.

⁷ Restructured standard advances are the standard advances of the bank that are restructured. These advances have higher probability of turning out bad but it gives banks the leeway to maintain lower provisions for these accounts and underestimate their total stressed assets.

⁸ Since all restructured advances are secured the provisioning requirement for such loans is 10%. In case of restructured standard advances RBI increased provisioning requirement from 2.75% in November 2012 to 5% in June 2013.

Capital Adequacy ratio (CAR) is an important indicator of financial soundness in banks and is given as the ratio of banks capital to its risk weighted assets. Ahamed & Mallick (2017) argue that through restructuring of assets most of the banks were able to understate NPAs reducing their amount of risk weighted assets in the balance sheet. Banks which are not capital constrained have lesser incentives to maintain higher loan loss provisions. Therefore, we postulate the third hypothesis as following:-

H3- Banks with higher CAR will have less incentive to restructure more accounts to understate NPAs and maintain lower provisions. Higher bank CAR would be inversely related to restructured assets in banks.

Indian banks with higher restructured assets could understate NPAs, maintain lower provision for restructured account and thus bolster their profitability. Ahamed & Mallick (2017) argue that by restructuring standard assets banks with lower provisions could understate NPAs and overstate profitability. On one hand banks that are less profitable or are incurring losses will have more incentive to restructure standard advances to overstate profitability against banks that are highly profitable who may not have the incentive to restructure standard advances to understate NPAs and overstate profitability.

H4- Higher profitability in banks is associated with less restructuring of assets.

Empirical Analysis

The research seeks to identify the drivers of restructured advances in Indian banks. The empirical analysis accounts for following issues: First, the study checks for the collinearity issues among explanatory variables to eliminate any multicollinearity concerns. Second, the study checks for unit root in the explanatory variables. Any presence of unit root in the variables would result in

spurious results. Third, the estimation adopts random effects model on the grounds that ownership an important explanatory dummy variable is time invariant and gets dropped out from the fixed effects analysis. Fourth, the results have been checked for robustness by using an alternate dependent variable.

Table 1 below presents the summary statistics of the explanatory variables which gives interesting insights. The explanatory variables have sufficient amount of variation needed for explaining the variation in restructured advances. NIM on an average is 2.91% for all the banks in the sample. There is considerable variation in CAR with the minimum and maximum values being 9.21 % to 277.45% respectively. The variable GNPA's stands at around 3.68% for the entire sample with the minimum and maximum value being 0 and 38.79% respectively.

<< Insert Table 1 Here>>

Multicollinearity among explanatory variables is an important issue which results in large standard errors in the empirical estimation. For identifying multicollinearity among explanatory variables the study uses the correlation table presented in table 2. From the correlation table it is quite evident that multicollinearity is not a concern for our estimation no evidence of high correlation is seen among these variables. We also checked for VIF values and it was less than 5 for all the explanatory variables eliminating any multicollinearity concerns.

<< Insert Table 2 Here>>

The unit root of all the variables was checked using Levin Lin Chu test which has been widely used in the analysis of unit root in balanced panel data set. This test has much greater power in estimating separate unit root test for each cross section element (Barbieri, 2006). The analysis was done to find out whether unit root among variables exist so that obtained results are not

spurious in nature. The results of unit root are presented in Table 3. The variables GNPA's showed the presence of unit root. The study took the difference of GNPA's to eliminate any unit root in the variable.

<< Insert Table 3 Here>>

The empirical analysis estimates the determinants of restructured advances using random effects model as the fixed effects fails to identify time constant dummy variable like Ownership. Also, the study doesn't identify any endogeneity concerns so random effects technique may be appropriate in this case. To avoid any ambiguity the study used Hausman test for the choice between fixed effects and random effects and found random effects as the preferred estimation technique. The results of Hausman test showed the p value of 0.98 confirming the use of random effects estimation.

The empirical estimation highlights that banks profitability in terms of net interest margin significantly influences restructured advances. The results suggest that banks that are less profitable or are incurring losses will have more incentive to restructure standard advances to overstate profitability against banks that are highly profitable who may not have the incentive to restructure standard advances to understate NPAs and overstate profitability. It can be argued that banks that are more profitable will restructure less of their advances to understate NPAs as they will have sufficient amount to maintain provisions for NPAs and still be profitable. Hence, profitable banks will restructure loans less as compared to banks that are less profitable who will restructure advances more to understate NPAs and bolster profitability. Particularly PSBs made use of CDR and restructuring mechanisms so as to upgrade non performing restructuring assets to standard category after a certain period and charge less to their net income (Ahamed &

Mallick, 2017). The results find support to Ahamed & Mallick (2017) study which found that PSBs with higher restructuring of assets could reduce their risk higher as compared to private and foreign banks. On the other hand against our hypothesis CAR and NPAs turned out to be insignificant highlighting that CAR and NPAs do not influence restructuring of assets in Indian banks.

<< Insert Table 4 Here>>

Robustness Checks

For robustness check of the results we estimated our main model results using alternate dependent variable. We estimated the model using the ratio of restructured standard advances to net advances. The findings are still robust with profitability and Ownership as significant variables implying profitability negatively influences restructured advances and public sector banks have restructured more to understate their NPAs. The results of robustness checks are presented in Table 5.

<< Insert Table 5 Here>>

Conclusion

The study contributes to the existing literature on regulatory forbearance and banking sector NPAs. This study examines the determinants of restructured assets in Indian banks particularly examining the impact of bank specific determinants on restructured assets in banks. The expected findings are that Profitability and Ownership significantly influence restructured assets in Indian banks. Banks that are more profitable restructure less of their advances as compared to low profitable banks that restructure more to understate NPAs and bolster their profitability.

Also, public sector banks undergo more restructuring of their assets relative to the private banks as they have been the ones who have used CDR mechanism more. Public sector banks with surging NPAs and low profitability restructure more to understate NPAs in their books and bolster profitability. Greater restructuring enables PSBs to hide NPAs under the carpet with the concession of regulatory forbearance.

Implications of the study

Indian banks have been resorting to large amount of stressed restructuring so as to sweep NPAs under the carpet. Restructured advances are very important component of galloping NPAs in Indian public banks. The existing literature on bank NPAs has failed to identify the determinants of restructured assets in banks. RBI and the Govt. of India should frame policies into account after accounting for factors that influence restructuring of assets by banks. The empirical estimation reveals that profitability and Ownership influences restructured assets in banks. RBI and policy makers should ensure that banks especially PSBs are profitable so that they don't resort to excessive restructuring of assets which should be curbed as it is resorted to understate NPAs and bolster profitability in these units. Greater profitability will ensure banks have adequate money to make provisions for NPAs as well as for lending. At the same time the restructuring practice in PSBs should be strictly dealt with as restructured assets are also problem loans. The RBI should strictly monitor restructuring practice in PSBs, along with increasing the provisioning requirement for these assets.

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Appendix

Table 1- Summary Statistics

Variable	Mean	Std. Dev.	Min	Max
RSA1	3.465425	19.32121	0	465.4563
NIM	2.912309	0.855989	0.23089	6.311533
CAR	15.17253	13.06783	9.21	277.45
Ownership	0.510204	0.500322	0	1
GNPAs	3.682369	3.498892	0	38.79644

Note: Number of observations = 588. RSA1- Restructured Standard Advances, NIM- Net Interest Margin, CAR-Capital Adequacy Ratio, Ownership dummy variable and GNPAs- Gross NPAs as a percentage of gross advances.

Table 2- Correlation Table

	NIM	CAR	Ownership	GNPAs
NIM	1			
CAR	0.1238	1		
Ownership	-0.4173	-0.2216	1	
GNPAs	-0.1724	0.1188	0.0134	1

Table 3- Unit root analysis using Levin Lin Chu Test

Variables	Adjusted t*	p value
RSA1	-6.37	0

NIM	-8.9	0
CAR	-6.36	0
GNPAs	2.79	0.99

**indicates significance at 5% and * indicates significance at 10%. Results are reported with robust standard errors.

Table4- Empirical results using Random Effects and Pooled OLS

Variables	Random Effects	OLS	Random Effects	OLS
NIM	-0.476* (0.06)	-0.42 (0.11)	-0.92** (0.02)	-0.85** (0.00)
CAR	0.009 (0.24)	0.009 (0.23)	0.004 (0.72)	0.004 (0.71)
Ownership	3.95** (0.03)	3.99** (0.02)	3.43** (0.01)	0.63** (0.01)
GNPAs	0.39 (0.36)	0.4 (0.34)	0.63 (0.35)	3.49 (0.33)
Year effects	Yes	Yes	No	No
F or Wald	397.15** (0.00)	19.44** (0.00)	22.08** (0.00)	8.91** (0.00)
Chi(P value)				
R Square	0.03	0.03	0.01	0.01

**indicates significance at 5% and * indicates significance at 10%. Results are reported with robust standard errors.

Table5- Robustness checks using alternate dependent variable

Variables	Random Effects	OLS	Random Effects	OLS
NIM	-0.47* (0.07)	-0.41 (0.13)	-0.94** (0.02)	-0.86** (0.00)
CAR	0.01 (0.18)	0.01 (0.16)	0.005 (0.66)	0.005 (0.65)
Ownership	4.14** (0.03)	4.2** (0.02)	3.59** (0.01)	3.66** (0.01)
GNPAs	0.41 (0.37)	0.41 (0.35)	0.65 (0.36)	0.65 (0.34)
Year effects	Yes	Yes	No	No
F or Wald	400.62**	19.15**	22.35**	9.30**
Chi (p value)	(0.00)	(0.00)	(0.00)	(0.00)

**indicates significance at 5% and * indicates significance at 10%. Results are reported with robust standard errors.